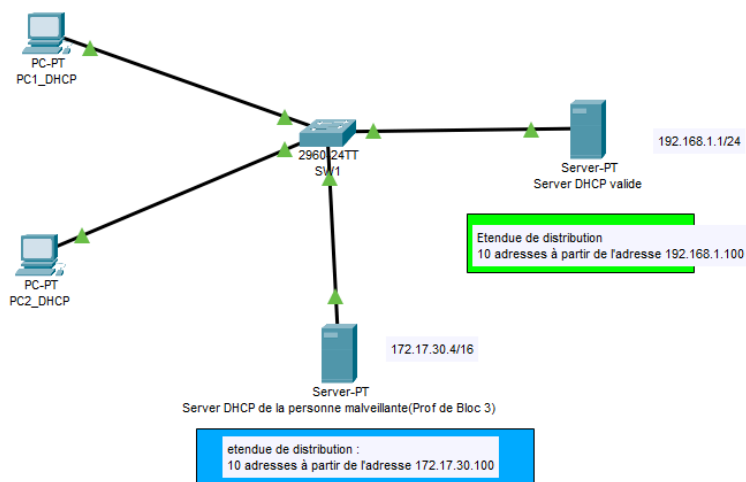


TP Attaque DHCP

Mise en place du schéma:.....	1
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Test du snooping.....	6

Mise en place du schéma:

D'abord, sur packet tracer, on refait comme sur le schéma donné.



On met les adresses IP sur les serveurs:
Pour le serveur valide

Server DHCP valide

Physical Config Services **Desktop** Programming Attributes

IP Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::206:2AFF:FE41:D4DE

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Pour le serveur malveillant

Server DHCP de la personne malveillante(Prof de Bloc 3)

Physical Config **Services** Desktop Programming Attributes

IP Configuration X

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 172.17.30.4

Subnet Mask: 255.255.0.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::260:2FFF:FE4B:51C4

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Maintenant on active le DHCP comme demandé respectivement sur les deux serveurs qu'on a modifié.

Server DHCP valide

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: server valide

Default Gateway: 192.168.1.254

DNS Server: 0.0.0.0

Start IP Address: 192 168 1 100

Subnet Mask: 255 255 255 0

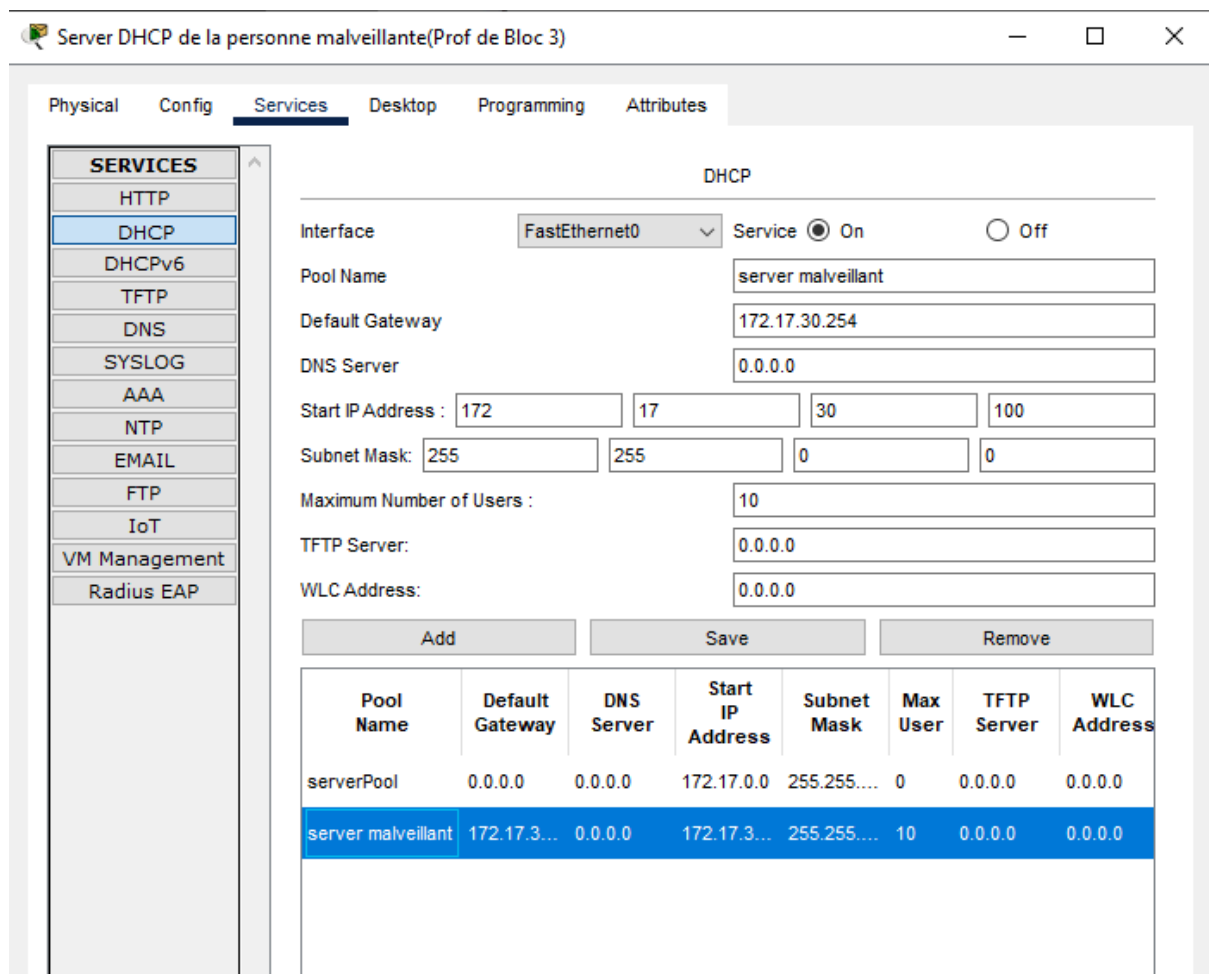
Maximum Number of Users: 10

TFTP Server: 0.0.0.0

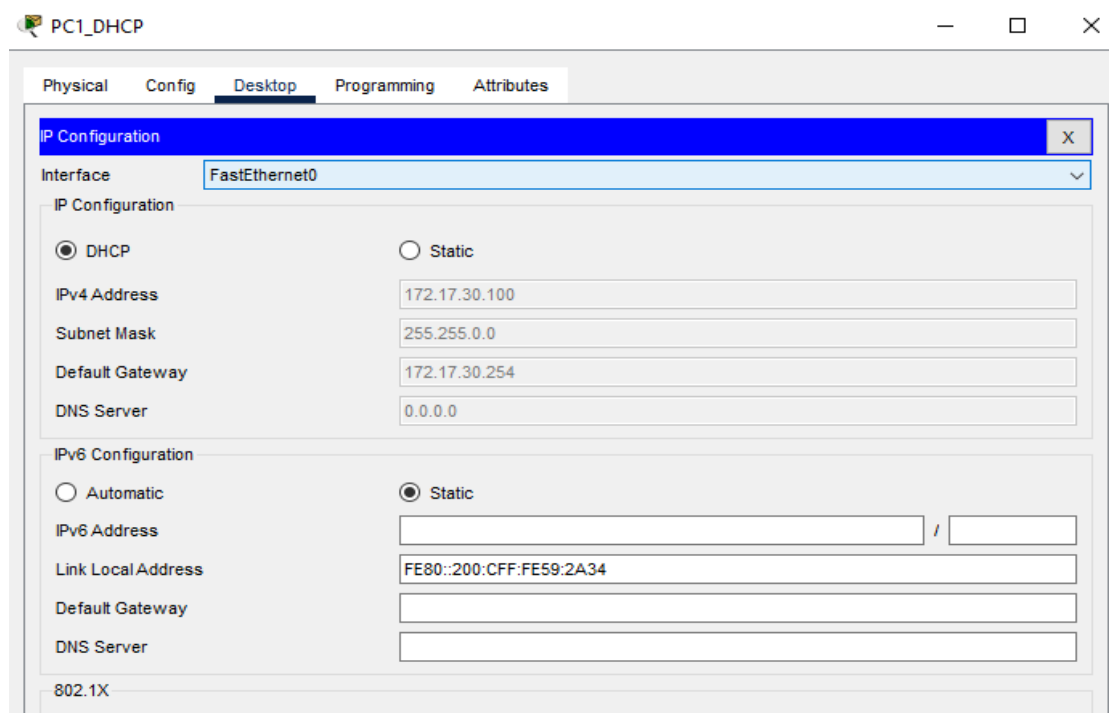
WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	0.0.0.0	0.0.0.0	192.168....	255.255....	0	0.0.0.0	0.0.0.0
server valide	192.168....	0.0.0.0	192.168....	255.255....	10	0.0.0.0	0.0.0.0



Il faut ensuite tester le DHCP sur les PC en l'activant dans leur configuration.
PC1 avec le serveur malveillant→



PC2 →

PC2_DHCP

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 172.17.30.101

Subnet Mask 255.255.0.0

Default Gateway 172.17.30.254

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:64FF:FEAD:37BD

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

PC1 avec le serveur valide:

PC1_DHCP

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.100

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.254

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::200:CFF:FE59:2A34

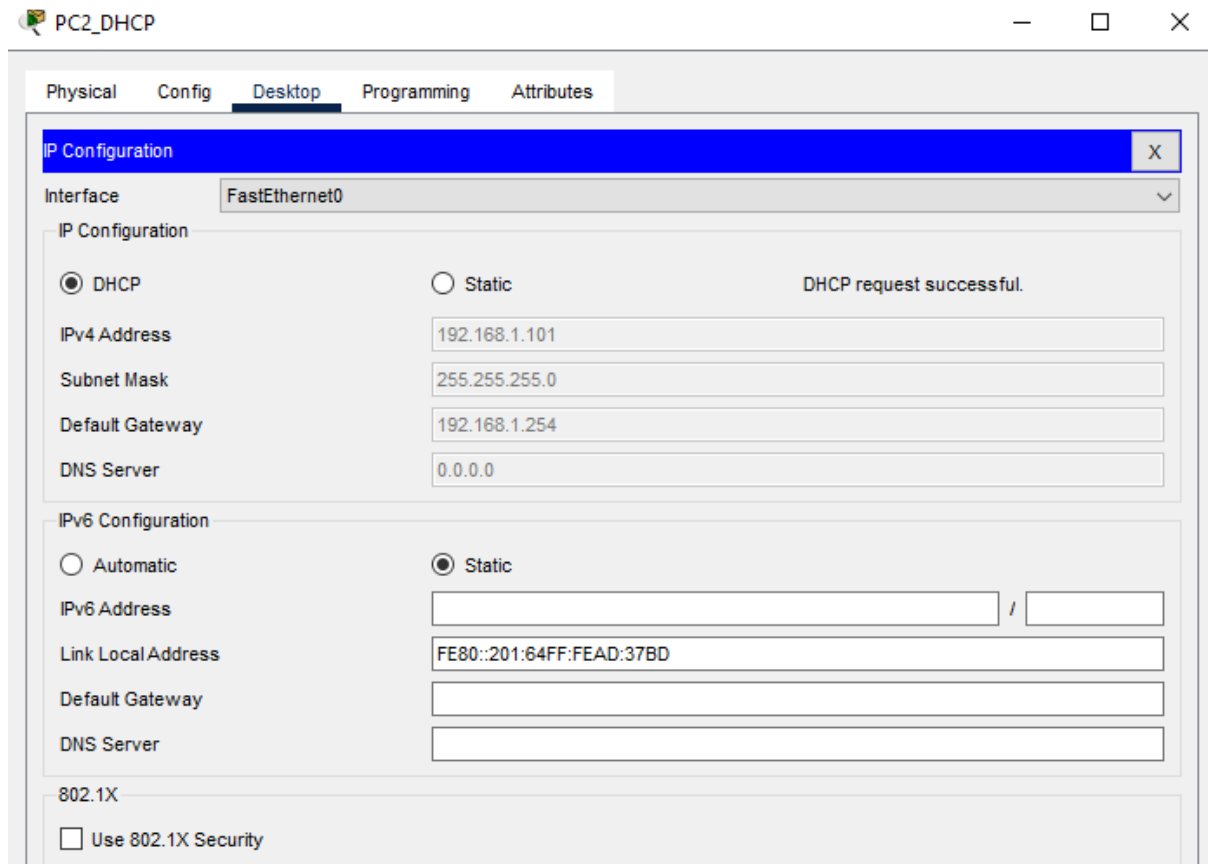
Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

PC2 →



Mise en place du DHCP snooping

On met en place le snooping d'abord en global (**ip dhcp snooping**). Ensuite on active l'option 82 qui permet de vérifier les paquets transmis. On sélectionne l'interface connecté au serveur valide(int fa0/1) avant de déclarer le port en tant qu'interface de confiance et de définir le maximum de requêtes que l'interface traitera par seconde(ici j'ai choisi 25).

```
SW1(config)#ip dhcp sno
SW1(config)#ip dhcp snooping
SW1(config)#ip dhcp snooping i
SW1(config)#ip dhcp snooping information o
SW1(config)#ip dhcp snooping information option
SW1(config)#int fa0/2
SW1(config-if)#ip dhcp sno
SW1(config-if)#ip dhcp snooping t
SW1(config-if)#ip dhcp snooping trust
SW1(config-if)#ip
SW1(config-if)#ip dhcp snoop
SW1(config-if)#ip dhcp snooping li
SW1(config-if)#ip dhcp snooping limit ra
SW1(config-if)#ip dhcp snooping limit rate 250
SW1(config-if)#
```

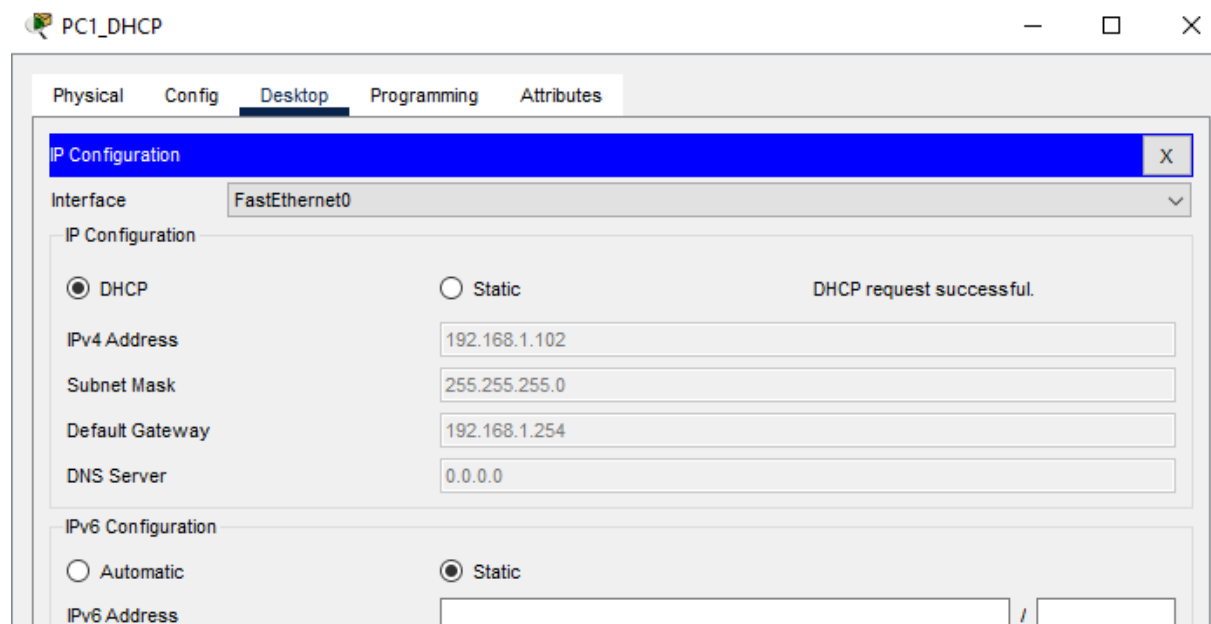
Maintenant, en exécutant la bonne commande, on peut voir les ports connectés au switch et ceux que le snooping autorise.

```
SW1#sh ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
none
Insertion of option 82 is enabled
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Interface                Trusted    Rate limit (pps)
-----
FastEthernet0/2          yes       250
FastEthernet0/1          no        unlimited
FastEthernet0/3          no        unlimited
FastEthernet0/4          no        unlimited
-----
```

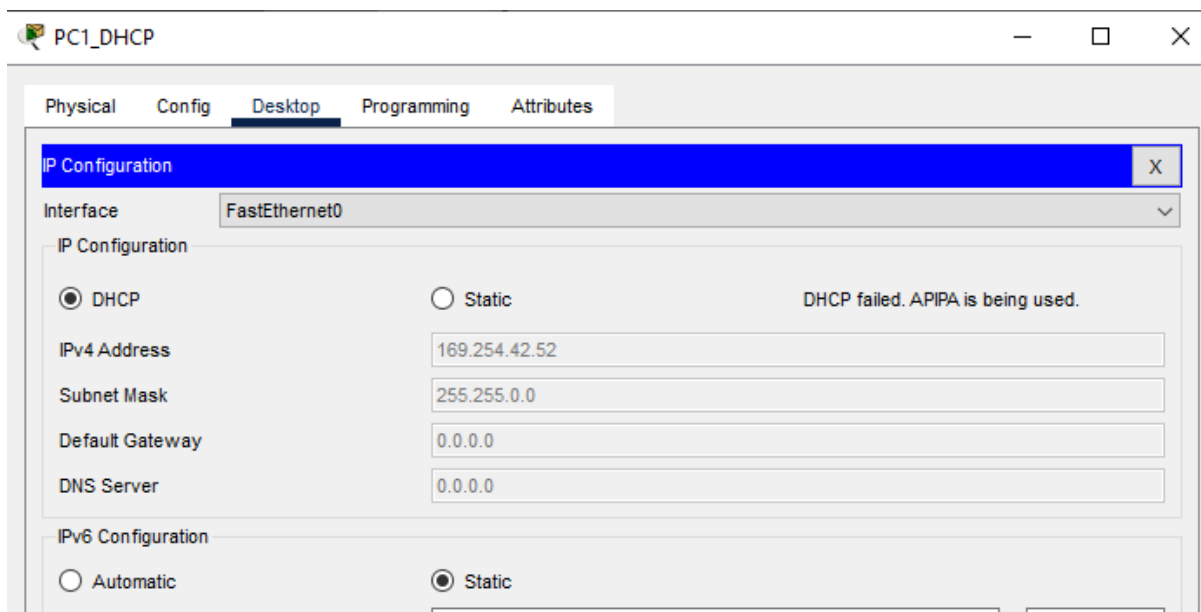
Test du snooping

Maintenant, le serveur valide continue de distribuer des baux mais lorsqu'on active seulement le serveur malveillant, plus rien n'est transmis.

D'abord en laissant le serveur valide allumé:



Et maintenant après l'avoir éteint et laissé allumé le malveillant



En mode simulation, lors de la demande DHCP, on peut voir que le switch refuse les baux du serveur malveillant

PDU Information at Device: PC2_DHCP

OSI Model Inbound PDU Details

At Device: PC2_DHCP
Source: SW1
Destination: STP Multicast Address

In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3	Layer3
Layer 2: IEEE 802.3 Header 00D0.BCB1.6204 >> 0180.C200.0000 LLC STP BPDU	Layer2
Layer 1: Port FastEthernet0	Layer1

1. The frame's destination MAC address matches the receiving port's MAC address, the broadcast address, or a multicast address.
2. The device does not have a service that accepts this frame. It drops the frame.

Challenge Me << Previous Layer Next Layer >>